JITHIN SHAJI

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github.com/jithin017 | https://jithin017.github.io/jithin-portfolio/

Career Objective

Results-driven Robotics and Automation Engineering student with proven expertise in intelligent systems, real-time control, and autonomous navigation. Lead technical development for Team Equinox, achieving All India Rank 7th in the aBAJA competition. Eager to apply advanced technical skills to contribute to innovative and future-ready robotics and AI engineering projects.

Education

Bachelor of Technology (B.Tech) in Robotics and Automation – Saintgits College of Engineering, Kottayam
CGPA: 7.43 | Expected Graduation: 2026

Higher Secondary Education (Class XII) – Good Shepherd Public School
Percentage: 85% | Year of Completion: 2022

Secondary Education (Class X) – Good Shepherd Public School
Percentage: 85% | Year of Completion: 2020

Skills

- **Programming Languages:** Python, C++, Data Structures & Algorithms, Object-Oriented Programming
- **Robotics & Motion Planning:** ROS 2, SLAM, Nav2, Gazebo, PID Control, Stanley Controller, EKF Sensor Fusion
- **CAD/CAM & Robotic Fabrication:** FreeCAD (custom build & workbench development), Toolpath Generation, STL/G-code Processing, Coordinate Transformation, Motion Planning for Robotic Manufacturing
- **Personal Attributes:** Leadership ,Teamwork ,Communication , Self-Learning Ability ,Time Management
- **Hardware & Embedded Systems:** NVIDIA Jetson Orin Nano, Raspberry Pi, ESP32, Arduino, LIDAR, IMU, Radar, CAN Bus, UART, I2C
- **Computer Vision & AI**: OpenCV, PyTorch, Object Detection & Classification, Dataset Creation & Labeling, Weights & Biases
- **Tools & Platforms**: Linux, Git, GitHub, SolidWorks, AutoCAD, Fusion 360,Kicad
- **Languages (Spoken/Written):** English, Malayalam

Professional & Technical Experience

Autonomous Systems - Team Equinox (ABAJA SAEINDIA 2025)
System Integration Lead | Saintgits College of Engineering, Kerala
Mar 2024 – Dec 2025

- Led technical development of a 30-member team, achieving All India Rank 7 in aBAJA SAEINDIA 2025, managing an $11,000 project budget and delivering the autonomous buggy two weeks ahead of schedule
- Developed a Level-2 autonomous buggy, integrating perception, path planning, CAN communication, and PID/Stanley controllers on NVIDIA Jetson Orin Nano under real-time constraints.
- Built traffic-light detection models using custom datasets and PyTorch, and implemented EKF-based radar fusion for obstacle detection, emergency braking, and safety overrides.

Volunteer Mentor | Dec 2025 – Present

- Provide technical guidance on ROS2 integration, autonomous navigation, and perception pipelines for the current team.

Internships

1. **Robot / Motion Control Programming Intern**
   Interface Design Associates Pvt. Ltd., Navi Mumbai
   Jan 2026 - April 2026

- Compiled and customized FreeCAD, developing UI and backend modules for a robotics/CNC workbench integrating 3D modelling and toolpath generation, improving development workflow efficiency by ~30%.
- Implemented toolpath and coordinate computation algorithms converting STL/G-code into robot motion commands, reducing manual motion-programming effort by ~40%.
- Developed ROS-based simulation pipelines to validate robotic 3D-printing trajectories and motion control logic, reducing debugging and deployment time by ~25%.

2. **Industrial Robot Programming Intern ,**
   PSG – FANUC Centre for Advanced CNC and Robotics (FANUC Authorised Training Centre)
   May 2024 – June 2024

- Programmed FANUC industrial robots using Teach Pendant (TP) to automate motion control and manufacturing tasks.
- Optimized pick-and-place automation routines, reducing cycle time by 12% with ±0.2 mm positioning repeatability..

3. **Artificial Intelligence & Machine Learning Intern, Innovate Intern**
   Jan 2024 – Feb 2024
   - Built a Flask-based Face Recognition Attendance System using OpenCV for real-time image processing and automated attendance logging.
   - Optimized the recognition pipeline, reducing processing latency by 80% while maintaining ~90% detection accuracy.
   - Implemented computer vision pipelines and data logging modules for efficient real-time attendance tracking.
4. **Embedded Systems and Robotics Intern, Techmaghi**
   June 2024 – July 2024
   - Designed a ROS 2-based sensor fusion node combining LIDAR and IMU data to generate a 3-D occupancy map for path planning.

Selected Technical Projects

1 . **Swarm Robot Development (ROS 2)** Completed

Role: ROS 2 Communication & Simulation Lead

- Developing a multi-robot system inspired by natural swarm behavior for coordinated warehouse automation tasks.
- Implementing ROS 2 framework for inter-robot communication, formation control, and obstacle avoidance algorithm.
- Conducting simulations and initial hardware trials to validate decentralized decision-making and scalability.

2 . **B.R.A.V.O (Hospital Assistive Robot with Smart Triage & Autonomous Navigation)** Ongoing

Role: ROS 2 Integration, Simulation & Raspberry Pi Lead

- Raspberry Pi–based robot for patient vitals collection and ROS 2–driven triage.
- Autonomous navigation using LIDAR, IMU, and mecanum wheels with room-specific smart action.
- Implemented autonomous navigation using LIDAR, IMU, and mecanum wheels, integrating ROS 2 for sensor fusion and decision-making.

3 . **VANIS Robot (Hack for Nothing Hackathon)**

Role:Lead Developer – Person Detection & Raspberry Pi Integration Completed

- Developed a Raspberry Pi–based robotic system using OpenCV for real-time human face detection.
- Integrated a phone camera (as an IP camera) along with ultrasonic and IR sensors for vision, obstacle avoidance, and autonomous hiding behavior.
- Implemented face-detection–based decision logic to trigger hiding maneuvers, achieving 92% detection accuracy with an average latency of 120 ms per frame.

Leadership & Achievements

- **BAJA SAEINDIA** National Competition (2025): Secured **All India Rank 7th** as System Integration Lead for Team Equinox.
- **Venue Coordination Head** – **11th National Level Technical Project Exhibition "Srishti 2025"** : Managed logistics, scheduling, and on-site operations for 300+ teams at a national technical exhibition.
- **Sub-Head** - Event "County Cricket," Nakshatra 2025 (National Level Techno-Cultural Fest) (2025): Coordinated registrations, scheduled 8 matches, and handled event operations at a national fest.
- NPTEL: Solid Mechanics (Elite) | 2024
- **Google Project Management** – Google (Coursera) | 2024
- Fundamentals on Robotics and Industrial Automation – L&T Edu Tech (Coursera) | 2024
- Executive Committee Member – College Magazine 2024 - Present : Contributed to planning, content review, and coordination for the college magazine's .
- Successfully completed the "Fundamentals of Python Programming" course, gaining a strong foundation in Python for automation and robotics applications.

Workshops

- Workshop on SELF DRIVING CAR conducted by **SAEINDIA Southern Section** at SAEISS, Guindy, Chennai.
- ROS 2 Training Participant , **PSG – FANUC Centre for Advanced CNC and Robotics (FANUC Authorised Training Centre)**
- ROS NOETIC in action workshop on 8th & 11th april 2024 conducted by iedc@saintgits under the banner of ipl 3.0
- Workshop on **PCB DESIGN** as part of THE ARC hosted by IEEE Student Branch Chapter
- Attended **PowerBi** Workshop conducted by OfficeMaster.

Membership in Professional Body

- **SAEINDIA: Member :** Member, Society of Automotive Engineers India
- **IEEE: Member :** Member of IEEE Student Branch
- **IETE: Member :** Member of Institution of Electronics and Telecommunication Engineers (IETE)